SMART FOOD WASTE REDISTRIBUTION SYSTEM

A PROJECT REPORT

Submitted by

MOHAMMED IBRAHIM M **2303811720521031**

MOHAN RAJ M **2303811720521033**

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K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(AUTONOMOUS)

SAMAYAPURAM - 621112

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K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(AUTONOMOUS)
SAMAYAPURAM - 621112

BONAFIDE CERTIFICATE

The work embodied in the present project report entitled **"SMART FOOD WASTE REDISTRIBUTION SYSTEM"** has been carried out by the students Mohammed Ibrahim M, Mohan Raj M. The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

Date of Viva Voce: ……………………

Mrs. J. MERCY GRACE
SUPERVISOR
Assistant Professor
Information Technology,
K. Ramakrishnan College of
Technology (Autonomous),
Samayapuram–621112.

Dr. C. SHYAMALA
HEAD OF THE DEPARTMENT
Associate Professor
Information Technology,
K. Ramakrishnan College of
Technology (Autonomous),
Samayapuram–621112.

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

A scalable and intelligent Smart Food Waste Redistribution System is designed to minimize food wastage and ensure efficient distribution of surplus food to those in need. The system follows a structured architecture that integrates donors, non-governmental organizations (NGOs), and volunteers through a centralized platform. It enables real-time food listing, smart matching based on location and expiry time, and secure delivery using OTP-based verification. The core functionalities include user management, food donation tracking, priority-based allocation, volunteer coordination, and real-time delivery monitoring. In addition, the system incorporates a sustainable waste management module that converts non-distributable food into useful by-products such as manure, thereby promoting environmental responsibility. The platform is designed to be modular, scalable, and efficient, with provisions for future enhancements such as AI-based waste prediction, food quality assessment, and route optimization. Overall, it provides a comprehensive solution for reducing food waste, improving resource utilization, and supporting social welfare initiatives.

Keywords: Smart Food Redistribution, Food Waste Management, NGO Integration, Volunteer Coordination,OTP Verification, Real-Time Tracking, Priority-Based Allocation, Sustainable Waste Processing, Manure Conversion, Web-Based Platform, Scalable System

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SIGNATURE

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INTRODUCTION

1.1 SMART FOOD WASTE REDISTRIBUTION SYSTEM

Food wastage has become a major global issue, with large quantities of edible food being discarded daily from households, restaurants, and events, while many people continue to face hunger. The absence of an efficient system to connect food donors with those in need leads to poor resource utilization. To address this problem, a Smart Food Waste Redistribution System is proposed, which uses digital technology to manage and redistribute surplus food effectively.

The system provides a centralized platform where donors such as hotels, households, and event organizers can upload details of excess food, including quantity, type, and expiry time. Non-governmental organizations (NGOs) and volunteers can access this information and accept requests for collection and distribution. By using location-based matching and priority allocation based on expiry time, the system ensures timely and efficient redistribution of food.

In addition to redistribution, the system also incorporates sustainable waste management practices. Food that is no longer suitable for consumption is identified and processed into useful by-products such as compost or manure. This approach helps in reducing environmental impact and ensures that no food goes completely to waste, promoting a zero-waste ecosystem.

The system also ensures secure and reliable operations through features such as OTP-based verification for delivery confirmation, role-based access control, and real-time tracking of food movement. These functionalities improve transparency and accountability among users. Overall, the Smart Food Waste Redistribution System provides a scalable, efficient, and socially responsible solution to reduce food waste and support community welfare.

1.2 NEED FOR FOOD WASTE MANAGEMENT

Food waste has become a major global concern, with a significant amount of edible food being discarded daily from households, restaurants, hotels, and large-scale events. At the same time, millions of people continue to suffer from hunger and malnutrition. This imbalance clearly indicates the need for an effective system to manage and redistribute surplus food efficiently.

The absence of proper coordination between food donors and recipients leads to large quantities of food being wasted unnecessarily. Many potential donors are willing to contribute excess food, but lack of awareness, time constraints, and absence of a proper platform prevent them from doing so. Hence, a structured and accessible system is required to connect donors with NGOs and volunteers in real time.

Food waste also has serious environmental consequences. When food is disposed of in landfills, it decomposes and produces methane gas, a major contributor to global warming. Additionally, wasting food means wasting valuable resources such as water, energy, and labor used during food production, processing, and transportation. Therefore, reducing food waste is essential for environmental sustainability.

An efficient food waste management system can help in minimizing waste by redistributing excess food to those in need. By using digital technologies such as real-time tracking, location-based matching, and expiry monitoring, the system ensures that food reaches beneficiaries quickly and safely before it becomes unusable.

In addition to redistribution, proper handling of non-edible or expired food is equally important. Instead of discarding such waste, it can be processed into compost or manure, which can be used in agriculture. This approach supports a circular economy where waste is converted into useful resources, reducing overall environmental impact.

Moreover, implementing a structured food waste management system can improve accountability and transparency in the redistribution process. Features such as real-time tracking, delivery verification, and user monitoring ensure that food reaches the intended beneficiaries without misuse. It also builds trust among donors, NGOs, and volunteers, encouraging more people to participate in the system.

1.3 ROLE OF NGOS AND VOLUNTEERS

Non-governmental organizations (NGOs) and volunteers play a vital role in the successful implementation of the Smart Food Waste Redistribution System. They act as the connecting link between food donors and beneficiaries, ensuring that surplus food is collected and distributed efficiently. Their involvement is essential for bridging the gap between food availability and demand.

NGOs are primarily responsible for managing and organizing the redistribution process. They receive notifications about available food, evaluate the requirements of different locations, and coordinate the allocation of resources accordingly. Their experience in handling social welfare activities enables them to ensure that food reaches the right people in a structured and effective manner.

Volunteers contribute by handling the operational aspects of the system. They are responsible for picking up food from donors and delivering it to NGOs or directly to beneficiaries. Their timely action is crucial in preventing food spoilage and ensuring quick distribution. The system assigns volunteers based on proximity and availability to optimize the delivery process.

To enhance coordination, the platform provides real-time notifications, task assignments, and tracking features. NGOs can monitor the progress of food collection and delivery, while volunteers receive updates about assigned tasks. This ensures smooth communication and minimizes delays in the redistribution process.

Security and accountability are maintained through features such as OTP-based verification during pickup and delivery. This ensures that only authorized volunteers handle the food and prevents misuse. Additionally, feedback and rating systems can be implemented to evaluate the performance of NGOs and volunteers, promoting transparency and trust.

Overall, NGOs and volunteers form the backbone of the system. Their active participation, coordination, and dedication ensure that surplus food is utilized effectively, reducing waste and supporting communities in need.

1.4 SCOPE OF THE PROJECT

The scope of the Smart Food Waste Redistribution System focuses on reducing food wastage by efficiently connecting food donors with NGOs and volunteers through a digital platform. The system aims to provide a structured solution for collecting, managing, and redistributing surplus food from various sources such as households, restaurants, hotels, and events to people in need.

The project covers the development of a centralized system that supports multiple user roles including donors, NGOs, and volunteers. Each user is provided with specific functionalities such as food listing, request management, task assignment, and delivery tracking. This ensures smooth coordination and effective utilization of available resources.

The system is designed to operate in real-time, enabling quick communication and faster decision-making. Features such as location-based matching and expiry-based prioritization ensure that food is distributed efficiently before it becomes unsuitable for consumption. This reduces delays and minimizes food spoilage.

In addition to redistribution, the project also includes waste management practices for handling non-edible or expired food. Such waste can be converted into compost or manure, contributing to environmental sustainability. This expands the system beyond redistribution into a complete food waste management solution.

The scope of the project also allows for scalability and future enhancements. Advanced features such as AI-based waste prediction, food quality analysis, and route optimization can be integrated to improve system performance. The platform can also be extended to support mobile applications and large-scale deployment.

Overall, the project aims to create a practical, efficient, and sustainable solution for managing food waste. By integrating technology, logistics, and community participation, the system has the potential to make a significant social and environmental impact.

1.5 APPLICATIONS

The Smart Food Waste Redistribution System can be widely applied in various sectors to reduce food wastage and improve food accessibility. One of the primary applications is in restaurants, hotels, and catering services, where excess food is generated on a daily basis. These establishments can use the system to quickly notify NGOs and volunteers, ensuring that surplus food is redistributed instead of being discarded.

The system is also highly useful in households and social events such as weddings, parties, and functions, where large quantities of food often remain unused. By using the platform, individuals can easily contribute leftover food to the system, enabling its redistribution to nearby communities in need.

Non-governmental organizations (NGOs) and charitable institutions can utilize the system to manage food collection and distribution more effectively. It helps them identify available food sources, track deliveries, and ensure that food reaches shelters, orphanages, and underprivileged areas in a timely manner.

The application is also beneficial for volunteers who wish to participate in social service activities. The system provides them with opportunities to contribute by handling food pickup and delivery tasks. It also helps in organizing volunteer efforts efficiently through task assignment and tracking features.

In addition, the system can be applied in environmental management by handling food waste responsibly. Non-edible or expired food can be processed into compost or manure, which can be used in agriculture. This reduces landfill waste and supports sustainable practices.

Overall, the Smart Food Waste Redistribution System has wide-ranging applications across social, environmental, and organizational domains.

CHAPTER 2
LITERATURE SURVEY

2.1 FOOD WASTE MANAGEMENT SYSTEMS AND CHALLENGES

AUTHOR: S. Thyberg & D. J. Tonjes

YEAR: 2016

DESCRIPTION:

This study focuses on the analysis and development of food waste management systems and highlights the major challenges involved in reducing food wastage. The authors explain how inefficient handling, lack of awareness, and poor coordination contribute significantly to global food waste. They emphasize the need for structured systems to manage surplus food effectively and reduce environmental impact.

The paper discusses advancements in food waste management practices, including improved waste tracking, resource recovery, and sustainable disposal methods. It also highlights the importance of integrating technology such as digital platforms and logistics systems to connect food donors with organizations that can redistribute food efficiently.

However, several challenges are identified, including technical issues such as system implementation, monitoring, and scalability, as well as social challenges like lack of public awareness and participation. The study also points out difficulties in ensuring food safety, maintaining quality during redistribution, and managing logistics efficiently.

Overall, the study concludes that effective food waste management requires a combination of technological solutions, policy support, and active community participation. Addressing these challenges is essential to minimize food waste and promote sustainability.

2.2 FOOD REDISTRIBUTION PLATFORMS IN SOCIAL WELFARE

AUTHOR: Paula Savary, Patrizia Beretta & Stephan Hellweg

YEAR: 2020

DESCRIPTION:

This study focuses on the role of food redistribution platforms in supporting social welfare and reducing food waste. The authors explain how surplus food from various sources such as households, retailers, and restaurants can be effectively redistributed to people in need through organized systems. These platforms act as intermediaries that connect food donors with charities and non-governmental organizations.

The paper highlights the importance of digital technologies in enabling efficient food redistribution. Mobile applications and web-based platforms are used to facilitate real-time communication, food listing, and coordination between donors and recipients. Features such as location-based matching and time-sensitive distribution help in ensuring that food is delivered before it spoils.

The study also discusses the social and environmental benefits of food redistribution systems. These platforms not only help in reducing food waste but also contribute to addressing food insecurity and supporting vulnerable communities. By improving resource utilization, they play a significant role in building sustainable societies.

However, the authors identify several challenges, including logistical constraints, food safety concerns, and the need for proper coordination among stakeholders. Ensuring timely pickup and delivery, maintaining food quality, and increasing participation from donors are some of the key issues highlighted.

Overall, the study concludes that food redistribution platforms are effective tools for social welfare, but their success depends on proper system design, user participation, and efficient coordination mechanisms.

2.3 ROLE OF NGOS IN FOOD DISTRIBUTION SYSTEMS

AUTHOR: R. Midgley

YEAR: 2014

DESCRIPTION:

This study focuses on the significant role of non-governmental organizations (NGOs) in food distribution systems, particularly in addressing hunger and food insecurity. The author explains how NGOs act as intermediaries between food donors and beneficiaries, ensuring that surplus food is collected and distributed efficiently to those in need. Their involvement is crucial in managing resources and reaching vulnerable communities.

The paper highlights that NGOs possess strong networks and local knowledge, which enable them to identify areas with high demand for food assistance. They coordinate with donors such as restaurants, supermarkets, and households to collect excess food and distribute it through shelters, food banks, and community centers. Their structured approach improves the effectiveness of food redistribution.

The study also discusses the operational challenges faced by NGOs, including limited resources, logistical constraints, and the need for proper coordination. Ensuring timely collection and delivery of food, maintaining quality standards, and managing volunteer participation are some of the key responsibilities handled by NGOs.

Additionally, the research emphasizes the importance of integrating technology into NGO operations. Digital platforms can enhance communication, improve tracking, and streamline the distribution process. This helps NGOs operate more efficiently and expand their reach.

Overall, the study concludes that NGOs play a vital role in food distribution systems, and their effectiveness can be further improved through better coordination, technological support, and increased community participation.

2.4 AI-BASED FOOD WASTE PREDICTION SYSTEMS

AUTHOR: N. Mehrabi, F. Morstatter, N. Saxena, K. Lerman & A. Galstyan

YEAR: 2021

DESCRIPTION:

This study explores the application of artificial intelligence techniques in predicting and managing food waste. The authors discuss how AI models can analyze historical data, consumption patterns, and environmental factors to estimate the amount of food likely to be wasted. Such predictive systems help organizations take preventive measures and improve resource planning.

The paper highlights the use of machine learning algorithms to identify trends in food usage across different sectors such as restaurants, households, and supply chains. By analyzing patterns, the system can provide recommendations such as adjusting food preparation quantities or initiating early redistribution, thereby reducing wastage.

The study also emphasizes the integration of AI with digital platforms to enhance decision-making. Real-time data processing and predictive analytics enable systems to prioritize food distribution based on urgency, location, and demand. This improves efficiency and ensures that surplus food is utilized effectively before it expires.

However, the authors identify challenges such as data availability, model accuracy, and system implementation complexity. Collecting reliable data and maintaining prediction accuracy are critical for the success of such systems. Additionally, the adoption of AI-based solutions requires technical infrastructure and user acceptance.

Overall, the study concludes that AI-based food waste prediction systems have significant potential in reducing waste and improving efficiency. When combined with proper data management and system integration, these technologies can contribute to sustainable and intelligent food management solutions.

2.5 SMART LOGISTICS IN FOOD DISTRIBUTION SYSTEMS

AUTHOR: M. Gevaers, E. Van de Voorde & T. Vanelslander

YEAR: 2011

DESCRIPTION:

This study focuses on the role of smart logistics in improving the efficiency of distribution systems, particularly in time-sensitive sectors such as food supply chains. The authors explain how advanced logistics strategies can optimize transportation, reduce delays, and ensure timely delivery of perishable goods. Efficient logistics plays a crucial role in minimizing food spoilage and waste during distribution.

The paper highlights the use of modern technologies such as GPS tracking, route optimization algorithms, and real-time communication systems in logistics operations. These technologies enable better coordination between different stakeholders, including suppliers, distributors, and delivery personnel. In food distribution systems, such features help ensure that surplus food is delivered quickly and safely to the intended recipients.

The study also discusses the importance of last-mile delivery optimization, which is critical in ensuring that food reaches its destination within a limited time frame. Proper route planning and resource allocation can significantly reduce transportation time and operational costs, improving overall system efficiency.

However, several challenges are identified, including high operational costs, infrastructure limitations, and the need for real-time data integration. Managing dynamic conditions such as traffic and availability of delivery personnel also adds complexity to logistics systems.

Overall, the study concludes that smart logistics is essential for efficient food distribution systems. By integrating advanced technologies and optimization techniques, logistics systems can significantly reduce delays, improve delivery performance, and support effective food redistribution.

2.6 SUSTAINABLE WASTE PROCESSING TECHNIQUES

AUTHOR: S. K. Papargyropoulou, R. Lozano

YEAR: 2014

DESCRIPTION:

This study focuses on sustainable waste processing techniques aimed at reducing the environmental impact of food waste. The authors explain the importance of managing food waste through structured approaches such as recycling, composting, and resource recovery. These methods help in converting waste into useful products instead of sending it to landfills.

The paper highlights various waste processing techniques including composting, anaerobic digestion, and bioenergy production. Composting converts organic waste into nutrient-rich manure, which can be used in agriculture. These techniques not only reduce waste volume but also contribute to environmental sustainability.

The study also emphasizes the concept of a waste hierarchy, where prevention and reuse are prioritized over disposal. By adopting sustainable processing methods, organizations can minimize environmental damage and promote efficient resource utilization.

However, the implementation of these techniques faces challenges such as lack of awareness, infrastructure limitations, and high initial costs. Proper segregation of waste and technological support are essential for effective processing.

Overall, the study concludes that sustainable waste processing techniques play a vital role in managing food waste. Integrating these methods into food management systems can significantly reduce environmental impact and support a circular economy.

2.7 COMPOSTING AND MANURE PRODUCTION METHODS

AUTHOR: C. A. Edwards & J. R. Burrows

YEAR: 1988

DESCRIPTION:

This study focuses on composting and manure production as effective methods for managing organic waste, including food waste. The authors explain how composting is a biological process in which microorganisms break down organic materials into nutrient-rich compost. This process plays a vital role in reducing waste and improving soil fertility.

The paper highlights different composting techniques such as aerobic composting, vermicomposting, and anaerobic decomposition. These methods help convert food waste into valuable organic manure that can be used in agriculture. Compost improves soil structure, enhances nutrient content, and supports sustainable farming practices.

The study also emphasizes the environmental benefits of composting, including reduction of landfill waste and decrease in greenhouse gas emissions. By converting food waste into useful products, composting contributes to a circular economy and reduces the negative impact on the environment.

However, the authors identify challenges such as the need for proper waste segregation, time required for decomposition, and maintenance of optimal conditions like temperature and moisture. Lack of awareness and infrastructure can also limit the adoption of composting practices.

Overall, the study concludes that composting and manure production are efficient and eco-friendly solutions for managing food waste. When integrated into food waste management systems, these methods can significantly reduce waste and promote environmental sustainability.

2.8 MOBILE-BASED FOOD DONATION APPLICATIONS

AUTHOR: S. Davies, A. Finlay, S. Blake & A. Brockett

YEAR: 2019

DESCRIPTION:

This study focuses on the use of mobile-based applications to facilitate food donation and reduce food waste. The authors explain how mobile platforms enable individuals, restaurants, and organizations to quickly share information about surplus food and connect with charities or volunteers for redistribution. These applications provide a convenient and accessible way to manage food donations in real time.

The paper highlights key features of mobile-based food donation systems, including user-friendly interfaces, location-based services, and instant notifications. These features help users easily register, upload food details, and coordinate pickup and delivery. Mobile applications improve efficiency by reducing response time and enabling faster communication between donors and recipients.

The study also discusses the social impact of such applications in addressing food insecurity and supporting community welfare. By making food donation simple and accessible, these platforms encourage more people to participate actively. They also help in building a network of donors and volunteers, strengthening the overall redistribution system.

However, the authors identify several challenges, including user adoption, data reliability, and maintaining food safety standards. Ensuring that food is handled properly and delivered on time remains a critical concern. Additionally, technical issues such as app performance and connectivity can affect system efficiency.

Overall, the study concludes that mobile-based food donation applications are effective tools for reducing food waste and improving food distribution. With proper design and user engagement, these systems can significantly enhance the efficiency and reach of food redistribution efforts.

2.9 REAL-TIME TRACKING IN SUPPLY CHAIN SYSTEMS

AUTHOR: M. Christopher

YEAR: 2016

DESCRIPTION:

This study focuses on the importance of real-time tracking in modern supply chain systems, particularly for time-sensitive goods such as food. The author explains how real-time visibility enables better monitoring of goods from source to destination, improving efficiency and reducing delays in the distribution process.

The paper highlights the use of technologies such as GPS, RFID, and IoT for tracking and monitoring shipments. These technologies provide accurate and up-to-date information about the location and status of goods, allowing organizations to make quick decisions and respond to changes effectively.

The study also emphasizes the role of real-time tracking in improving transparency and accountability within supply chains. Stakeholders can monitor the movement of goods, ensuring that deliveries are completed as expected. In the context of food distribution, this helps prevent spoilage by ensuring timely delivery.

However, challenges such as high implementation costs, data management issues, and the need for reliable connectivity are identified. Integrating multiple tracking technologies and maintaining system accuracy can also be complex.

Overall, the study concludes that real-time tracking is a critical component of efficient supply chain management. When applied to food redistribution systems, it can significantly improve delivery efficiency, reduce waste, and enhance system reliability.

2.10 SECURE VERIFICATION SYSTEMS (OTP/AUTHENTICATION)

AUTHOR: N. Daswani, C. Kern & A. Kesavan

YEAR: 2007

DESCRIPTION:

This study focuses on secure authentication mechanisms used in modern digital systems to ensure safe and reliable user interactions. The authors explain how authentication techniques such as One-Time Passwords (OTP), multi-factor authentication, and secure session management help prevent unauthorized access and ensure data security.

The paper highlights the importance of OTP-based verification systems, where a unique code is generated and sent to the user for validating transactions or actions. This method enhances security by ensuring that only authorized users can complete sensitive operations such as login, data access, or service requests. OTP systems are widely used due to their simplicity and effectiveness.

The study also discusses the role of secure authentication in maintaining system integrity and user trust. By implementing strong verification mechanisms, systems can prevent fraud, misuse, and unauthorized activities. In applications involving multiple users and transactions, such as service platforms, authentication becomes a critical component.

However, challenges such as network dependency, delay in OTP delivery, and user inconvenience are identified. Additionally, ensuring secure transmission of authentication data and protecting against cyber threats require robust system design.

Overall, the study concludes that secure verification systems are essential for building reliable digital platforms. Integrating authentication mechanisms like OTP enhances security, improves user confidence, and ensures safe operation of the system.

CHAPTER 3
PROJECT METHODOLOGY

3.1 PROPOSED WORK

The proposed system aims to design and develop a comprehensive Smart Food Waste Redistribution System that addresses the growing issue of food wastage and hunger. The system integrates modern digital technologies to create a platform that connects food donors, non-governmental organizations (NGOs), and volunteers in a structured and efficient manner. It ensures that surplus food is not wasted but instead redirected to those who need it the most.

The system is designed as a centralized platform where multiple users can interact and collaborate. Donors such as restaurants, households, and event organizers can easily register and upload details of excess food. NGOs can monitor available food resources and manage distribution, while volunteers are responsible for collecting and delivering food. This structured coordination improves efficiency and reduces delays in the redistribution process.

A key component of the system is the food listing module, which allows donors to provide detailed information about available food. This includes food type, quantity, preparation time, and expiry time. By maintaining accurate and complete information, the system ensures that NGOs can make informed decisions regarding food collection and distribution.

The system incorporates an expiry tracking mechanism that prioritizes food distribution based on urgency. Food items that are closer to their expiry time are given higher priority, ensuring that they are delivered quickly. This feature significantly reduces food spoilage and improves the effectiveness of the redistribution process.

Another important feature is the smart matching system, which connects donors with nearby NGOs and available volunteers. The matching process considers factors such as location, food quantity, urgency, and volunteer availability. This ensures optimal allocation of resources and minimizes transportation time.

The system also includes a real-time notification mechanism that keeps all users informed about updates and activities. Donors receive confirmations when their food is accepted, NGOs get alerts for new food listings, and volunteers are notified about assigned tasks. This continuous communication enhances coordination and system efficiency.

A dedicated delivery management module is implemented to handle the logistics of food transportation. Volunteers are assigned tasks based on proximity and availability. The system tracks the entire delivery process, from food pickup to final distribution, ensuring transparency and accountability.

To ensure secure operations, the system incorporates OTP-based verification during both pickup and delivery stages. This mechanism confirms that the correct volunteer is handling the food and that the delivery is completed successfully. It helps in preventing misuse and builds trust among users.

The platform also integrates a reward-based system to motivate users. Donors and volunteers are awarded points, badges, or recognition for their contributions. This gamification approach encourages continuous participation and increases user engagement within the system.

In addition to redistribution, the system addresses the issue of non-edible or expired food through a waste management module. Such food is categorized and processed into compost or manure. This promotes environmental sustainability and ensures that no food is completely wasted.

From a technical perspective, the backend of the system is designed to handle multiple operations efficiently. It manages user authentication, data processing, and system logic. The database stores information such as user profiles, food listings, delivery records, and reward points in a structured format.

The system follows a modular architecture, allowing easy maintenance and future scalability. Each module operates independently while being integrated into the overall system. This design enables the addition of new features without affecting existing functionalities.

The frontend of the system is developed with a focus on usability and responsiveness. It provides a simple and intuitive interface for users to interact with the platform. Features such as dashboards, tracking interfaces, and notification panels enhance the overall user experience.

The system can be deployed on cloud-based infrastructure to ensure scalability, reliability, and high availability. Cloud deployment also supports data backup, load balancing, and improved system performance. This makes the platform suitable for large-scale implementation.

Future enhancements of the system may include advanced features such as AI-based food waste prediction, food quality analysis using image processing, and route optimization for faster delivery. These features can further improve system efficiency and effectiveness.

Overall, the proposed work focuses on creating a smart, scalable, and sustainable solution for food waste management. By integrating technology, logistics, and community participation, the system aims to reduce food wastage, support needy populations, and promote environmental responsibility.

3.2 BLOCK DIAGRAM

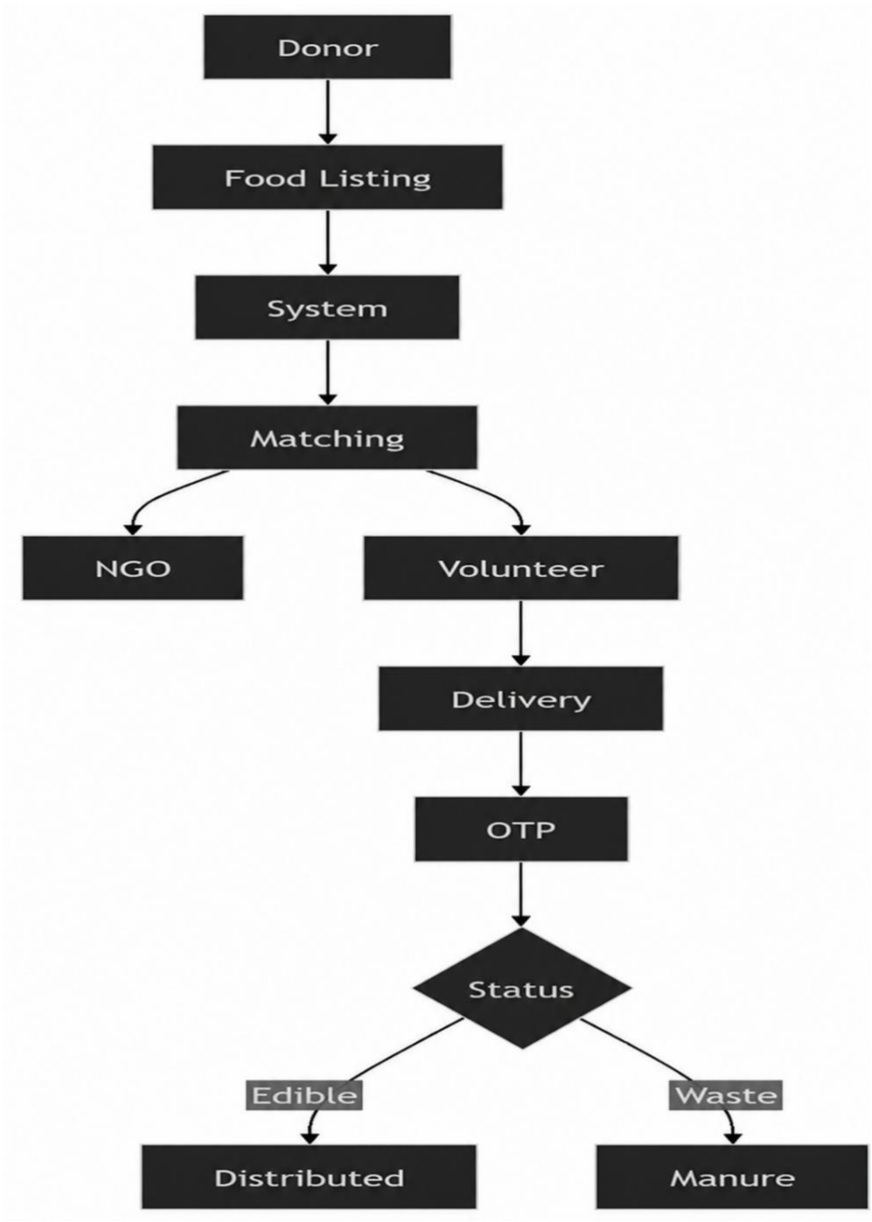

Figure 3.1 Smart Food Waste Redistribution System – Block Diagram

CHAPTER 4
MODULE DESCRIPTION

4.1 USER MANAGEMENT (DONOR/NGO/VOLUNTEER)

The User Management module is a fundamental component of the Smart Food Waste Redistribution System, responsible for handling different types of users such as donors, NGOs, and volunteers. This module ensures that each user can register, log in, and access functionalities based on their specific roles within the system.

The system allows users to create accounts by providing basic details such as name, contact information, and role selection. Based on the selected role, users are granted access to specific features. Donors can upload food details, NGOs can manage requests and distribution, and volunteers can handle pickup and delivery tasks.

Authentication mechanisms are implemented to verify user identity and protect sensitive data. Secure login processes, including password protection and session management, are used to prevent unauthorized access. Additionally, OTP-based verification can be incorporated to enhance security during critical actions such as delivery confirmation.

The module also maintains user profiles, where details such as activity history, contributions, and participation levels are stored. This information can be used to track user engagement and performance within the system. It also helps in identifying active users and improving overall system efficiency.

To improve reliability and trust, the system can include a rating and feedback mechanism. Users such as NGOs and volunteers can be evaluated based on their performance, responsiveness, and reliability. This helps in maintaining quality standards and encourages responsible participation.

4.2 FOOD LISTING AND EXPIRY TRACKING MODULE

The Food Listing and Expiry Tracking module is a core component of the Smart Food Waste Redistribution System. It enables donors such as households, restaurants, hotels, and event organizers to upload details of surplus food into the system. This module ensures that all available food is properly recorded and made accessible for redistribution.

Donors can provide detailed information about the food, including type, quantity, preparation time, and location. This information helps NGOs and volunteers understand the nature of the food and plan the distribution process effectively. A user-friendly interface is provided to make the food listing process simple and efficient.

One of the key features of this module is expiry tracking. Donors are required to specify the expiry time or duration for which the food remains safe for consumption. The system continuously monitors this information to ensure that food is distributed before it becomes unsuitable.

The module incorporates a priority mechanism based on expiry time. Food items that are closer to their expiry are given higher priority during allocation and distribution. This helps in reducing food spoilage and ensures that maximum food is utilized effectively.

Real-time updates are also supported in this module. Once food is listed, NGOs and volunteers are notified immediately, enabling quick response and faster coordination. This reduces delays in the redistribution process and improves overall system efficiency.

Overall, the Food Listing and Expiry Tracking module plays a vital role in managing surplus food efficiently. By combining accurate data entry, real-time monitoring, and priority-based allocation, it ensures timely redistribution and minimizes food wastage.

4.3 SMART MATCHING & ALLOCATION MODULE

The Smart Matching and Allocation module is a key component of the Smart Food Waste Redistribution System that ensures efficient connection between food donors, NGOs, and volunteers. It is responsible for identifying the most suitable NGO and available volunteer for each food donation request, enabling quick and effective redistribution.

This module uses parameters such as location, food quantity, expiry time, and user availability to perform matching. By analyzing these factors, the system selects the nearest NGO and assigns a volunteer who can handle the pickup and delivery within the required time frame. This reduces delays and ensures that food reaches beneficiaries before it spoils.

The allocation process is designed to be dynamic and responsive. When a donor uploads food details, the system immediately processes the information and generates matching results. NGOs receive notifications and can accept the request, after which the system assigns a volunteer automatically based on availability and proximity.

Priority-based allocation is another important feature of this module. Food items that are closer to their expiry time are given higher priority, ensuring urgent distribution. This approach minimizes food wastage and improves the overall efficiency of the system.

The module also supports real-time updates and tracking. Any changes in availability or status are reflected instantly, allowing users to stay informed about the progress of matching and allocation. This enhances coordination among all participants involved in the process.

Overall, the Smart Matching and Allocation module improves the effectiveness of food redistribution by ensuring optimal resource utilization. By combining location-based matching, priority handling, and real-time coordination, it plays a crucial role in reducing food waste and supporting timely distribution.

4.4 VOLUNTEER COORDINATION & OTP VERIFICATION

The Volunteer Coordination and OTP Verification module manages the assignment and activities of volunteers involved in the food redistribution process. It ensures that food pickup and delivery are carried out efficiently by assigning tasks to volunteers based on their availability and location. This helps in reducing delays and improving the overall delivery process.

Volunteers receive real-time notifications when a food pickup request is assigned to them. They can view details such as donor location, food type, quantity, and delivery destination. This information helps them plan the pickup and delivery process effectively, ensuring timely distribution.

The system assigns volunteers dynamically by considering factors such as proximity to the donor, availability, and workload. This optimized allocation ensures that the nearest and most suitable volunteer is selected, reducing travel time and improving efficiency.

A key feature of this module is OTP-based verification, which is used during both pickup and delivery stages. When a volunteer reaches the donor location, an OTP is generated and verified to confirm the authenticity of the pickup. Similarly, another OTP is used at the delivery point to confirm successful completion.

This verification mechanism enhances security and prevents unauthorized handling of food. It ensures that only assigned volunteers can perform the tasks and that the delivery is completed correctly. It also builds trust among donors, NGOs, and beneficiaries.

Overall, the Volunteer Coordination and OTP Verification module ensures smooth, secure, and efficient execution of food delivery operations. By combining task assignment, real-time communication, and secure verification, it plays a vital role in maintaining system reliability and accountability.

4.5 WASTE MANAGEMENT (MANURE CONVERSION)

The Waste Management module is an essential part of the Smart Food Waste Redistribution System, designed to handle food that cannot be redistributed for consumption. It ensures that non-edible or expired food is managed in an environmentally responsible manner instead of being discarded as waste.

This module classifies food into different categories such as edible, partially usable, and non-edible. While edible food is redistributed, non-edible or expired food is directed to the waste processing unit. This classification helps in proper handling and reduces unnecessary disposal.

The system incorporates processes such as composting and manure conversion to transform organic food waste into useful by-products. Through biological decomposition, food waste is converted into nutrient-rich compost that can be used in agriculture. This promotes sustainable practices and reduces environmental impact.

The module can also support coordination with local composting centers or agricultural units where the processed waste can be utilized. This ensures that waste is not only reduced but also converted into valuable resources that benefit the environment.

Proper tracking and monitoring of waste processing activities are also included in this module. The system records the quantity of waste processed and the amount converted into manure, which can be displayed in reports and dashboards.

Overall, the Waste Management module contributes to building a zero-waste ecosystem. By converting unused food into useful products, it supports environmental sustainability, reduces landfill waste, and enhances the overall effectiveness of the system.

CHAPTER 5

CONCLUSION AND FUTURE ENHANCEMENT

5.1 CONCLUSION

The proposed Smart Food Waste Redistribution System presents a comprehensive, scalable, and user-centric platform that effectively integrates food donation, real-time tracking, volunteer coordination, and sustainable waste management into a unified system. By providing structured access for donors, NGOs, and volunteers, the system ensures efficient collection and redistribution of surplus food, significantly reducing wastage and supporting communities in need.

The system enhances operational efficiency through features such as food listing with expiry tracking, smart matching based on location and urgency, and real-time notifications. The inclusion of OTP-based verification ensures secure and reliable delivery, while tracking mechanisms improve transparency and accountability throughout the redistribution process. These features collectively contribute to a well-organized and trustworthy platform.

Furthermore, the integration of a waste management module ensures that non-edible or expired food is not discarded but converted into useful by-products such as compost or manure. This promotes environmental sustainability and supports the concept of a zero-waste ecosystem. The reward-based mechanism also encourages active participation from donors and volunteers, improving user engagement and consistency.

Overall, the proposed system addresses critical challenges such as food wastage, inefficient distribution, and lack of coordination among stakeholders. By combining technology, logistics, and community participation, the system provides an effective and socially responsible solution that benefits both society and the environment.

5.2 FUTURE ENHANCEMENT

The Smart Food Waste Redistribution System can be further enhanced by incorporating advanced technologies and features to improve its efficiency, scalability, and user experience. One of the key enhancements is the integration of artificial intelligence and machine learning techniques to predict food waste patterns. These technologies can analyze historical data and user behavior to provide recommendations such as optimized food preparation and early redistribution.

Advanced features such as image-based food quality detection can be integrated to assess the freshness and safety of food before redistribution. This will improve food safety standards and reduce the risk of distributing spoiled food. Additionally, route optimization algorithms can be implemented to identify the fastest and most efficient delivery paths, minimizing delays and ensuring timely distribution.

The system can also be enhanced by developing a dedicated mobile application for Android and iOS platforms. This will improve accessibility and allow users to interact with the system anytime and anywhere. Features such as push notifications, GPS tracking, and real-time updates can further enhance user experience and system responsiveness.

From a technical perspective, the platform can be deployed on cloud infrastructure to support large-scale operations, ensure high availability, and provide data backup. Integration with external systems such as government food programs and large-scale NGOs can expand the reach and impact of the system.

Additional enhancements such as advanced analytics dashboards can provide insights into food distribution patterns, user participation, and system performance. These insights can help administrators make informed decisions and improve system efficiency over time.

APPENDIX A
SOURCE CODE

1. server.ts

```
import express from "express";
import { createServer as createViteServer } from "vite";
import path from "path";
import { fileURLToPath } from "url";
import Database from "better-sqlite3";

const __filename = fileURLToPath(import.meta.url);
const __dirname = path.dirname(__filename);

// --- SQLite Database Initialization ---
const db = new Database("foodbridge.db");

// Initialize Tables
db.exec(`
  CREATE TABLE IF NOT EXISTS users (
    uid TEXT PRIMARY KEY,
    email TEXT UNIQUE,
    name TEXT,
    role TEXT,
    isVerified INTEGER DEFAULT 1,
    createdAt TEXT
  );

  CREATE TABLE IF NOT EXISTS listings (
    id TEXT PRIMARY KEY,
    donorId TEXT,
    donorName TEXT,
```

```
    foodName TEXT,
    quantity TEXT,
    expiryTime TEXT,
    location TEXT,
    status TEXT,
    ngoId TEXT,
    ngoName TEXT,
    volunteerId TEXT,
    volunteerName TEXT,
    deliveryOtp TEXT,
    createdAt TEXT,
    FOREIGN KEY(donorId) REFERENCES users(uid)
  );
`);

async function startServer() {
  const app = express();
  const PORT = 3000;

  app.use(express.json());

  // --- Auth Routes ---
  app.post("/api/auth/register", (req, res) => {
    const { email, name, role } = req.body;
    try {
      const uid = Math.random().toString(36).substring(7);
      const createdAt = new Date().toISOString();
      const insert = db.prepare("INSERT INTO users (uid, email, name, role, createdAt)
VALUES (?, ?, ?, ?, ?)");
      insert.run(uid, email, name, role, createdAt);
```

```
    const user = db.prepare("SELECT * FROM users WHERE uid = ?").get(uid);
    res.json(user);
  } catch (err: any) {
    if (err.message.includes("UNIQUE constraint failed")) {
      return res.status(400).json({ error: "Email already registered" });
    }
    res.status(500).json({ error: err.message });
  }
});

app.post("/api/auth/login", (req, res) => {
  const { email } = req.body;
  const user = db.prepare("SELECT * FROM users WHERE email = ?").get(email);
  if (!user) return res.status(401).json({ error: "Account not found" });
  res.json(user);
});

app.get("/api/auth/users", (req, res) => {
  // Basic protection: usually checking a session or token would happen here
  const rows = db.prepare("SELECT * FROM users ORDER BY createdAt
DESC").all();
  res.json(rows);
});

// --- Listing Routes ---
app.get("/api/listings", (req, res) => {
  const rows = db.prepare("SELECT * FROM listings ORDER BY createdAt
DESC").all();
  res.json(rows);
});
```

```
  app.post("/api/listings", (req, res) => {
    const { donorId, donorName, foodName, quantity, expiryTime, location, status } =
req.body;
    const id = Math.random().toString(36).substring(7);
    const createdAt = new Date().toISOString();

    const insert = db.prepare(`
      INSERT INTO listings (id, donorId, donorName, foodName, quantity, expiryTime,
location, status, createdAt)
      VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?)
    `);
    insert.run(id, donorId, donorName, foodName, quantity, expiryTime, location, status,
createdAt);

    res.json({ id, ...req.body, createdAt });
  });

  app.patch("/api/listings/:id", (req, res) => {
    const { id } = req.params;
    const current = db.prepare("SELECT * FROM listings WHERE id = ?").get(id) as any;
    if (!current) return res.status(404).json({ error: "Not found" });

    const updateBody = req.body;

    // Logic for Delivery OTP
    if (current.status === 'AVAILABLE' && updateBody.status === 'CLAIMED') {
      updateBody.deliveryOtp = Math.floor(100000 + Math.random() * 900000).toString();
    }

    // Logic for Delivery Verification
    if (updateBody.status === 'DELIVERED') {
```

```
      if (!updateBody.otpVerify || updateBody.otpVerify !== current.deliveryOtp) {
        return res.status(403).json({ error: "Invalid Delivery OTP. Please ask the NGO for
the correct 6-digit code." });
      }
      delete updateBody.otpVerify;
    }

    const finalKeys = Object.keys(updateBody);
    const finalValues = Object.values(updateBody);

    const sets = finalKeys.map(k => `${k} = ?`).join(", ");
    const update = db.prepare(`UPDATE listings SET ${sets} WHERE id = ?`);
    update.run(...finalValues, id);

    const updated = db.prepare("SELECT * FROM listings WHERE id = ?").get(id);
    res.json(updated);
  });

  app.delete("/api/listings/:id", (req, res) => {
    db.prepare("DELETE FROM listings WHERE id = ?").run(req.params.id);
    res.json({ success: true });
  });

  // --- Background Task: Automatic Expiry to Manure ---
  setInterval(() => {
    const now = new Date().toISOString();
    const result = db.prepare(`
      UPDATE listings
      SET status = 'EXPIRED_TO_MANURE'
      WHERE status IN ('AVAILABLE', 'CLAIMED')
      AND expiryTime < ?
```

```
    `).run(now);

    if (result.changes > 0) {
      console.log(`[System] ${result.changes} items expired and sent to manure
processing.`);
    }
  }, 10000); // Check every 10 seconds for demo responsiveness

  // Vite middleware for development and static serving for production
  if (process.env.NODE_ENV !== "production") {
    const vite = await createViteServer({
      server: { middlewareMode: true },
      appType: "spa",
    });
    app.use(vite.middlewares);
  } else {
    const distPath = path.join(process.cwd(), 'dist');
    app.use(express.static(distPath));
    app.get('*', (req, res) => {
      res.sendFile(path.join(distPath, 'index.html'));
    });
  }

  app.listen(PORT, "0.0.0.0", () => {
    console.log(`Server running on http://localhost:${PORT}`);
  });
}

startServer();

{
```

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  "messagingSenderId": "757007379451",
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```


```
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    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>My Google AI Studio App</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.tsx"></script>
  </body>
</html>
```


```
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  "description": "A smart food waste redistribution platform connecting donors, NGOs, and volunteers to minimize food waste and support those in need.",
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  "clean": "rm -rf dist",
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    "allowJs": true,
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```

```
        "./*"
      ]
    },
    "allowImportingTsExtensions": true,
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}
```

```
import tailwindcss from '@tailwindcss/vite';
import react from '@vitejs/plugin-react';
import path from 'path';
import {defineConfig, loadEnv} from 'vite';

export default defineConfig(({mode}) => {
  const env = loadEnv(mode, '.', '');
  return {
    plugins: [react(), tailwindcss()],
    define: {
      'process.env.GEMINI_API_KEY': JSON.stringify(env.GEMINI_API_KEY),
    },
    resolve: {
      alias: {
        '@': path.resolve(__dirname, '.'),
      },
    },
    server: {
      // HMR is disabled in AI Studio via DISABLE_HMR env var.
      // Do not modifyâ□”file watching is disabled to prevent flickering during agent edits.
      hmr: process.env.DISABLE_HMR !== 'true',
    },
  };
```

```
});

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APPENDEX B

SCREENSHOTS

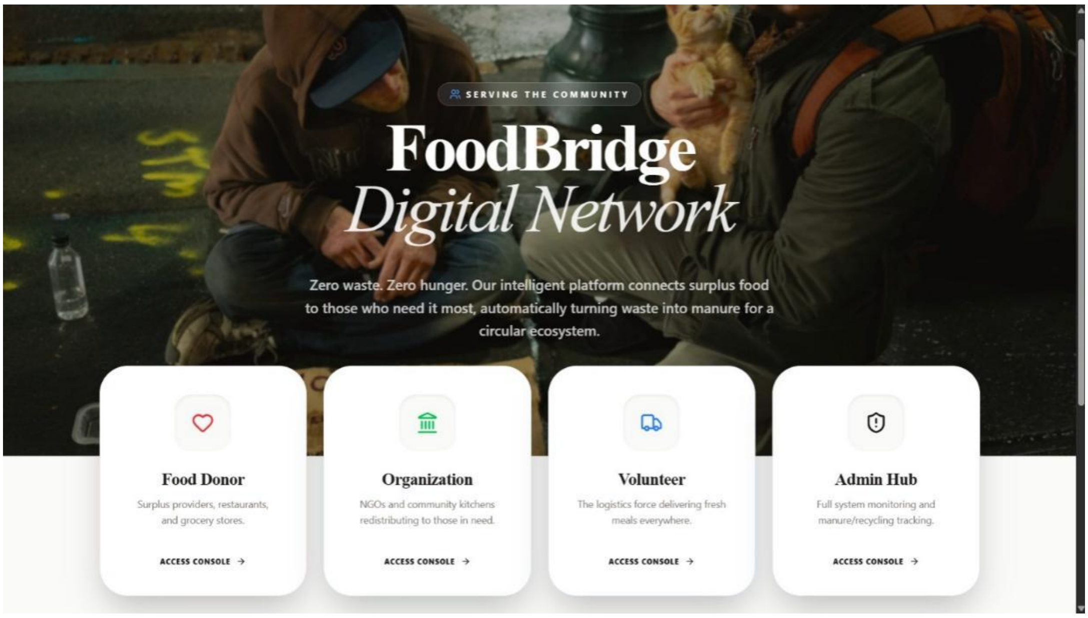

Figure B.1 Home interface

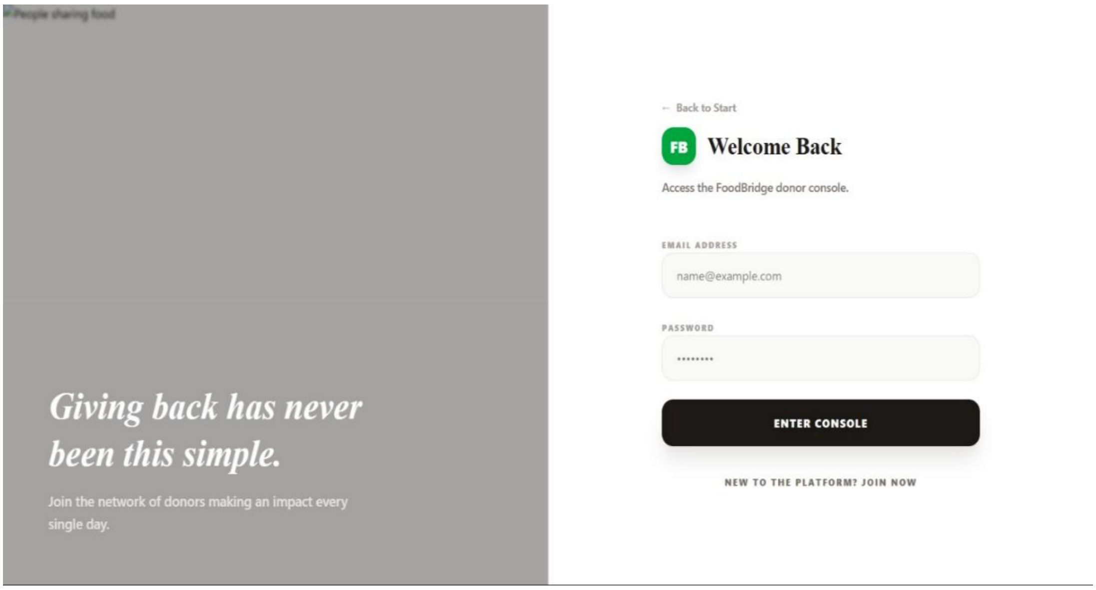

Figure B.2 Donor login page

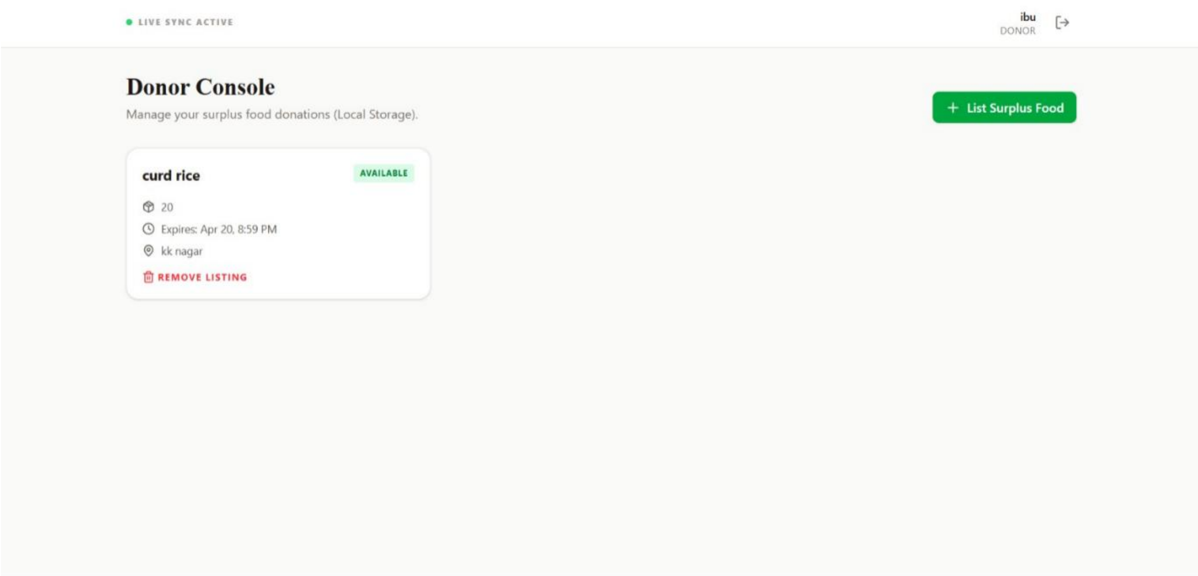

Figure B.3 Dashboard

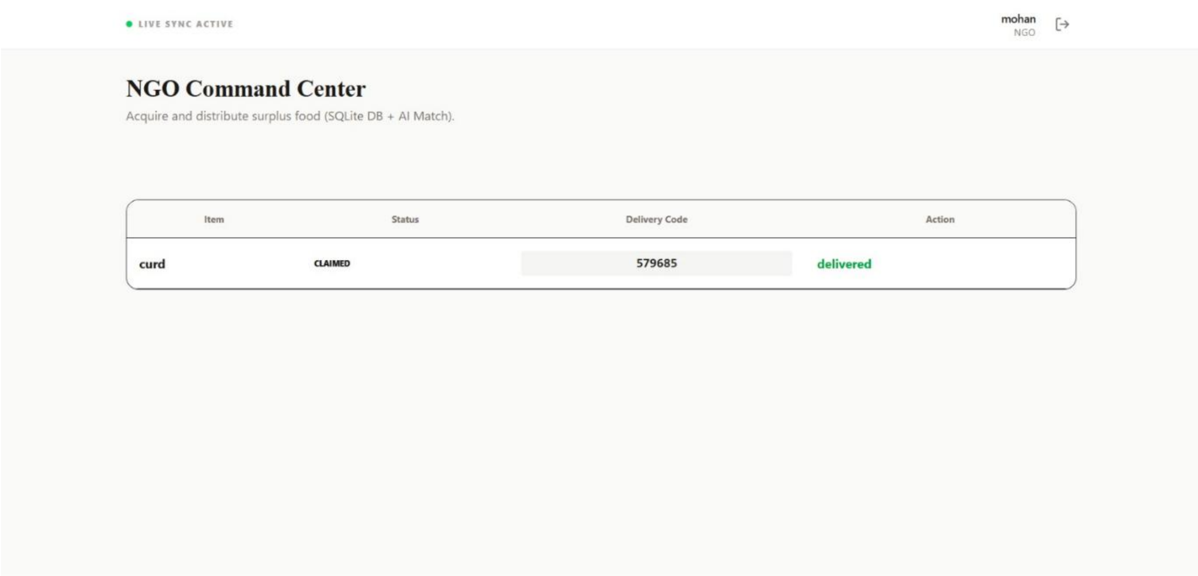

Figure B.4 Food allocation system

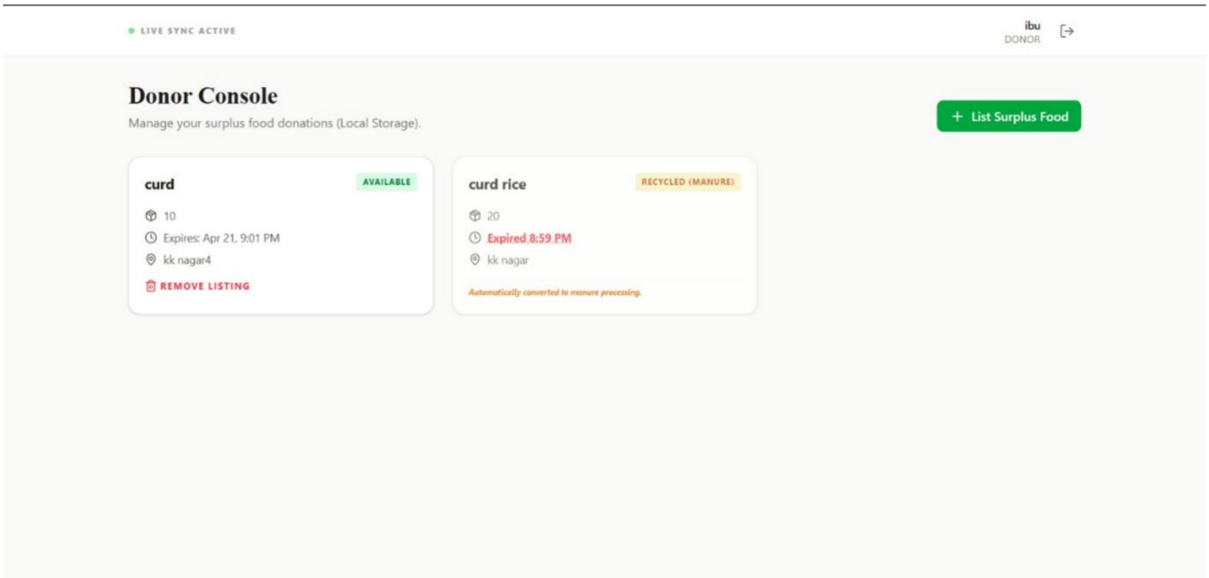

Figure B.5 Expiry and recycling status

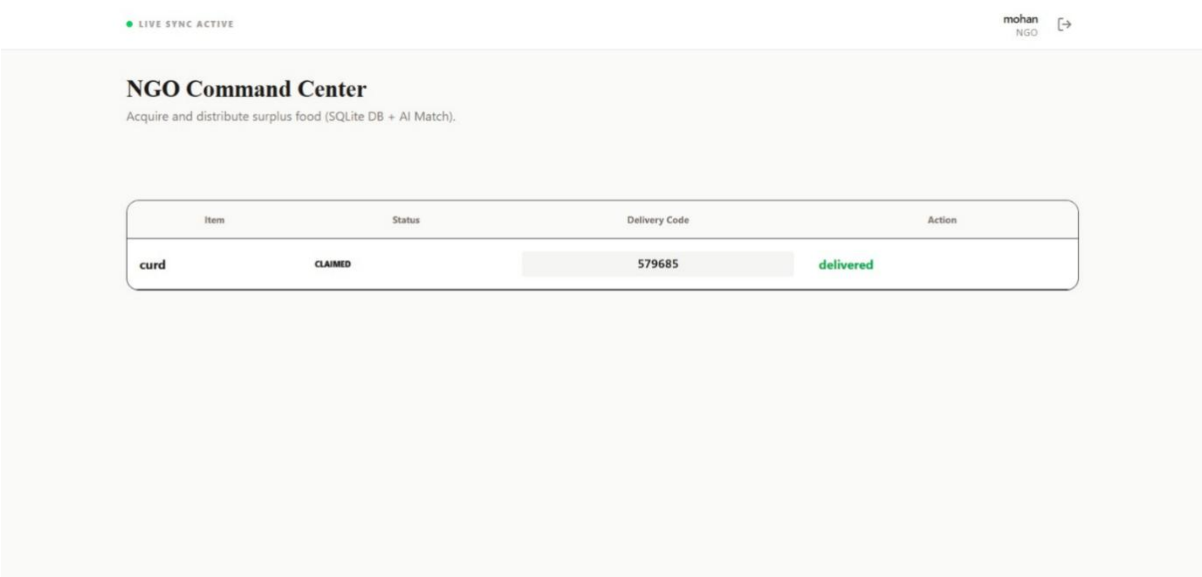

Figure B.6 Delivery assignment interface

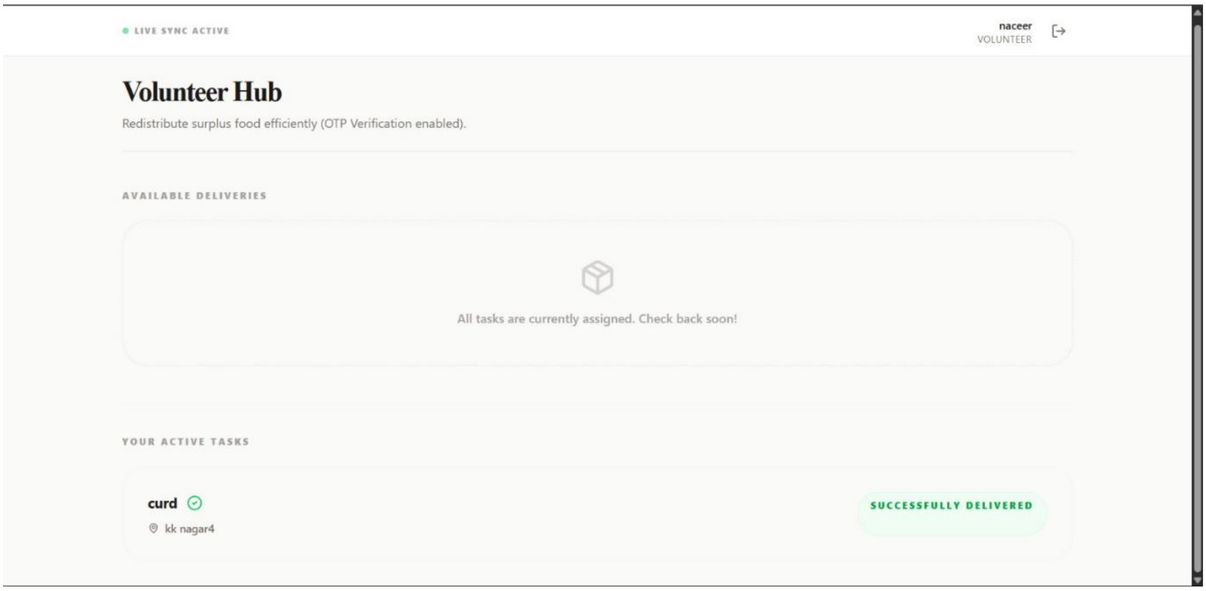

Figure B.7 Delivery completion status

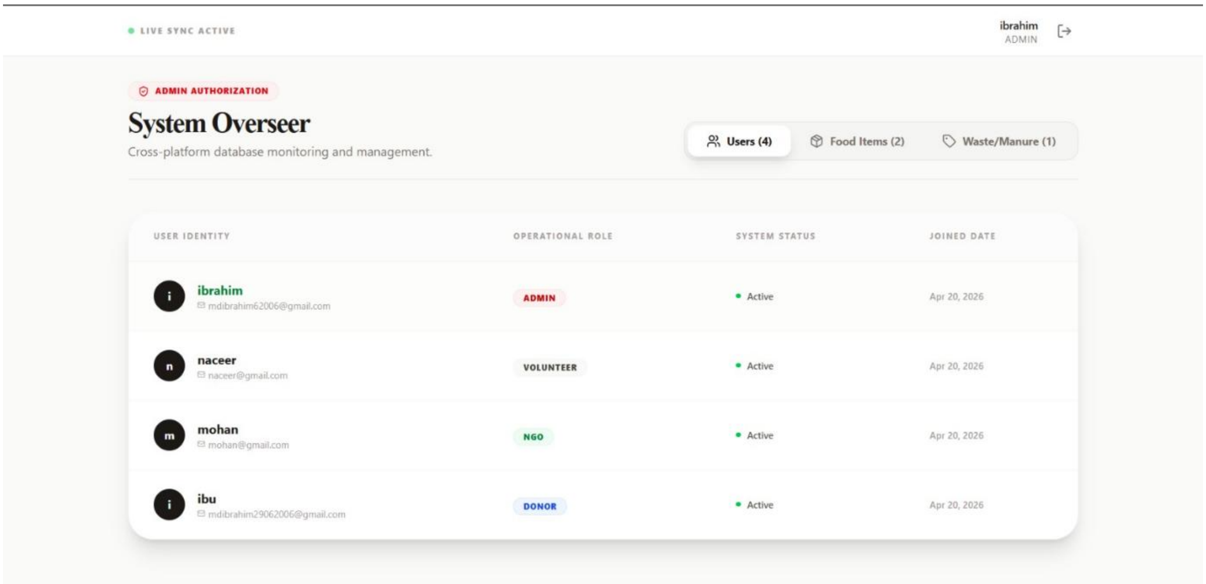

Figure B.8 System user management

REFERENCES

1. traceability systems in food supply chains, enabling better tracking, transparency, and safety management from production to consumption.
2. Buzby, J. C., et al. - 2014. *'The Estimated Amount, Value, and Calories of Postharvest Food Losses.'* This report analyzes food losses after harvest, estimating their economic value and caloric impact while highlighting the need for improved storage and distribution systems.
3. European Commission - 2020. *'Food Waste Prevention in the European Union.'* This work focuses on strategies and policies implemented across the EU to reduce food waste and improve sustainability in food management systems.
4. FAO - 2011. *'Global Food Losses and Food Waste – Extent, Causes and Prevention.'* This report provides a comprehensive analysis of global food waste, its causes at different stages, and possible prevention methods.
5. Gustavsson, J., et al. - 2011. *'Global Food Losses and Food Waste.'* This study presents global estimates of food loss and waste, emphasizing the environmental and economic impacts of inefficient food supply chains.
6. Kummu, M., et al. - 2012. *'Lost Food, Wasted Resources: Global Food Supply Chain Losses.'* This research highlights the resource wastage associated with food loss, including water, energy, and land, stressing the importance of efficient distribution systems.
7. Papargyropoulou, E., et al. - 2014. *'The Food Waste Hierarchy as a Framework for Prevention.'* This paper introduces a structured hierarchy for food waste management, prioritizing prevention, reuse, recycling, and disposal.